Jennifer Jiang-Kells

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Experience

dotimplement ai — Feb 2025 - Present
Founder / Open Source Maintainer — *London, UK*

- Founded and maintain HealthChain, an open-source Python SDK for healthcare AI deployment with native FHIR support and EHR connectivity; active pilot engagement with NHS research hospitals
- 199 Github stars, 16 contributors, 37 forks in 8 months; featured in TLDR AI (900K+ developers) newsletter; presented at Google NHS AI Hackathon, NHS Python Open Source Community Conference

University College London / University College Hospitals NHS Trust — Sep 2021 - Jan 2025
Senior Software Developer / Honorary Researcher — *London, UK*

- Built MiADE, a clinically evaluated NLP system for real-time diagnosis extraction at the point-of-care, deployed in live clinical use at two major NHS hospitals via HL7 CDA messaging with Epic
- Led end-to-end technical development including data pipelines, model training, deployment, and EHR API integration; achieved 83% precision / 85% recall for clinical diagnosis extraction (SNOMED CT) and reduced clinician structured data entry time by 89% in simulation testing
- Co-secured £605k EPSRC/UKRI research grant to optimize clinical data extraction using LLMs; led LLM engineering (miade-llm), contributed to study design, grant application, and multi-site model evaluation across UCLH and GOSH

LV8 Sports Ltd. — Sep 2019 – Aug 2021
Software Developer — *London, UK*

- Early engineer on GrowFootball iOS app — built custom ML architectures (object detection, pose estimation, AR) and end-to-end biomechanics data pipeline, achieving 20% model performance improvement

University College London — Mar 2018 – Sep 2018
Computational Neuroscience Research Assistant — *London, UK*

- Worked in DeepMind-funded research on the neural basis of spatial navigation in Prof Caswell Barry's lab
- Modelled irregular grid cells using maximum-likelihood decoding of simulated Poisson neural spiking dynamics

Publications

Jiang-Kells, J. et al. (2025). Design and implementation of a natural language processing system at the point of care: MiADE. *BMC Medical Informatics and Decision Making*, *25*, 365.

Education

Imperial College London — 2018-2019
MSc Computing Science

University College London — 2016-2017
MSc Clinical Neuroscience

University College London — 2013-2016
BSc Human Sciences

Technical Skills

Languages: Proficient - Python, C++, SQL / Experience in - Java, R, MATLAB
Technologies: SpaCy, TensorFlow, PyTorch, Sklearn, HuggingFace, Langchain, Flask, FastAPI, Docker, Nginx, Azure, Git, Streamlit, Vue.js, Vercel
Clinical Standards: FHIR, HL7, SNOMED CT, OMOP
Concepts: Natural Language Processing, Artificial Intelligence, Machine Learning, Neural Networks, Model Evaluation, Data Analyses, Computer Vision, Transformers, Large Language Models, API, Agile Methodology, Microservices, Cloud Computing